
MP1: Familiarization with the Tools

Due Time: Refer to the course web-site

Objectives

Verify that we are able to:

- Work with Unity Shaders and Materials
- Create and customize a shader
- Send customization data to the Shaders during runtime
- Name and clean up a Unity project for submission

Approach

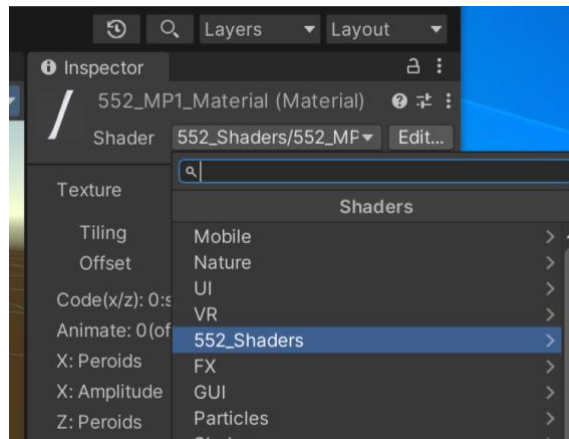
You will demonstrate your ability to work with the Unity editor by creating a project with a sphere and a capsule. Both of these primitive should be rendered with the 552_MP1_Material.

CSS_MP1_Material must:

- **Shader:** A simple shader that output a constant color
- **Color:** The color must be under user control

Shader:

- File name: 552_Shader.shader
- Category: 552_Shaders
 - Shaders are organized into *categories* where in the Inspector window it is easier to find the shader you want to associate with a material. The following screenshot shows your shader should be found under the **552_Shaders** category.



- Vertex Shader:
 - Program Name: ***VertexProgram***
 - Name of datatype when program is called: ***DataFromVertex***
 - Name of datatype for the output of the program: ***DataForFragmentShader***
 - You must perform the proper transformations by calling the HLSL function ***mul()***. In other words, you ***cannot*** call the ***UnityObjectToClipPos()*** macro.

- Fragment Shader:
 - Program Name: *FragmentProgram*
 - Name of datatype when program is called: *DataForFragmentShader*
 - Name of datatype for the output of the program: *OutputFromFragmentShader*
 - Output the constant color specified by the user.

Credit Distribution

	Credits
1 Settings in Unity	
Material name: <i>552_MP1_Material</i>	1
Shader file name: <i>552_Shader</i>	1
Customized UI category: <i>552_Shaders</i>	2
2 Shader customization	
Working with no useless line of code	10
Vertex shader program name	1
Fragment shader program name	1
3 Vertex shader	
Explicit mul() to transform vertices	5
Datatype name for input and output	2
4 Fragment shader	
Able to customize resulting color	5
Datatype name for input and output	2
5 Scene Setup	
Two primitives with your material	5
6 Submission	
Proper project name	2
Four folders in submission	3
Total	40

This programming assignment counts 3% towards your final grade for this class.